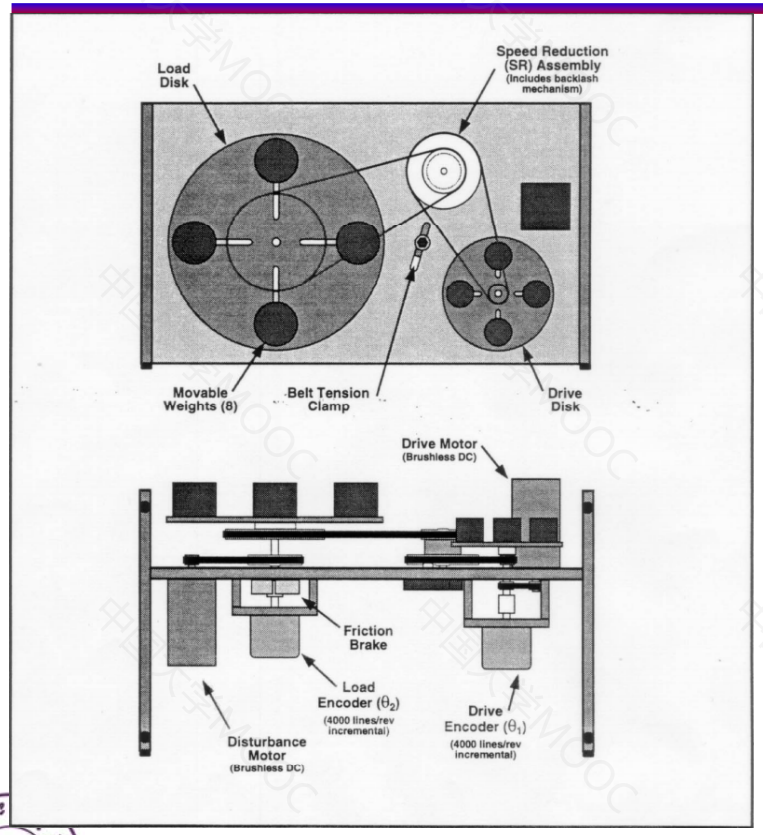
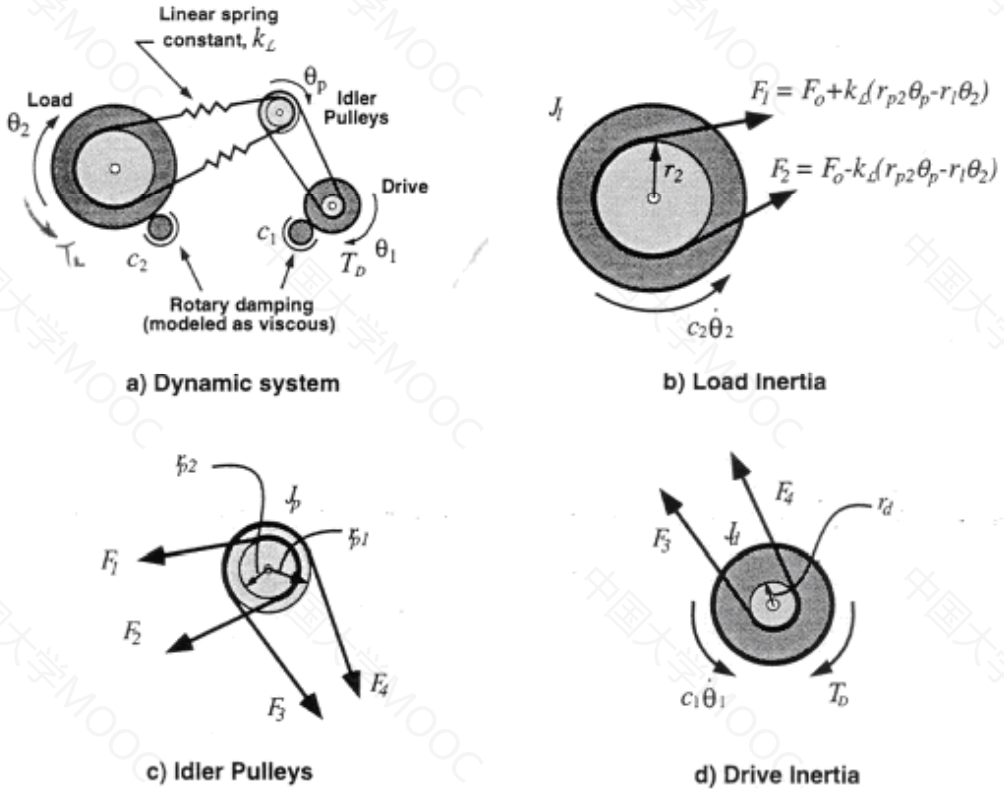
System Identification via Frequency Response

(通过频域响应的系统辨识)

Ex. Modeling and Identification of ECP Emulator





Ex. Modeling and Identification of ECP Emulator

for the drive disk $J_D\ddot{\theta}_1 = T_D - c_1\dot{\theta}_1 - T_{PD}$

$$T_{PD} = r_d(F_4 - F_3)$$

for the load disk $J_L\ddot{\theta}_2 = T_{PL} - c_2\dot{\theta}_2 - T_L$

$$T_{PL} = r_L(F_1 - F_2)$$

for the pulleys $J_P\ddot{\theta}_P = T_{P1} - T_{P2}$

$$T_{P1} = r_{p1}(F_4 - F_3)$$

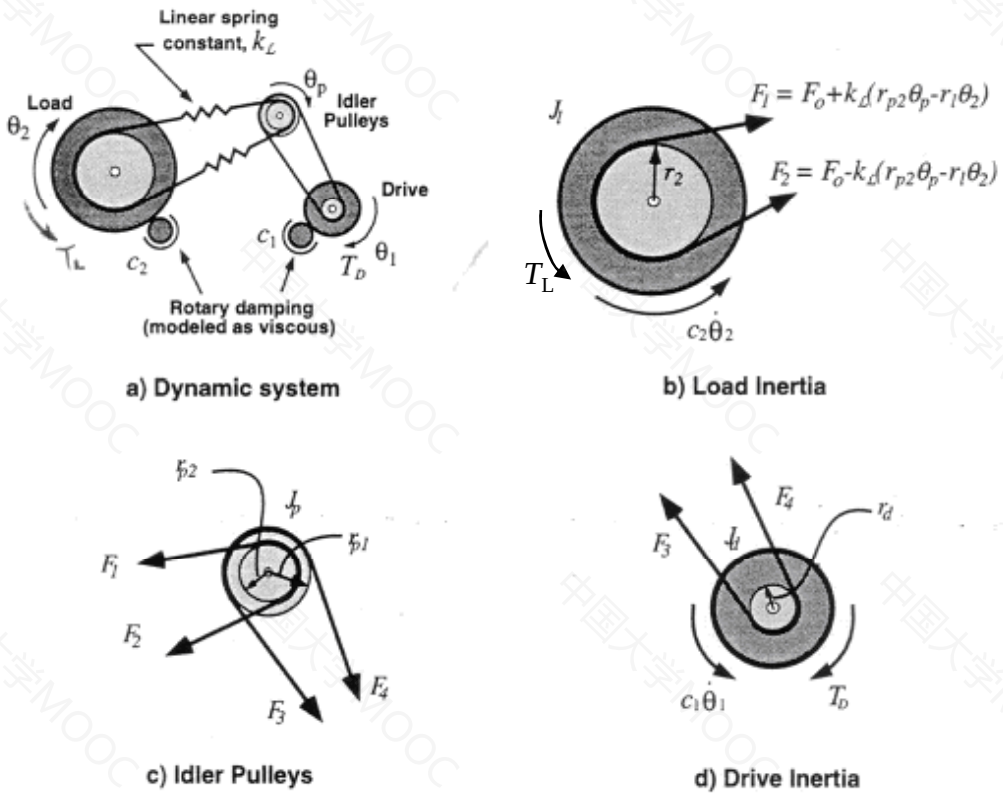
$$T_{P2} = r_{p2}(F_2 - F_1)$$

$$F_1 = F_{10} + k_1(r_{p2}\theta_P - r_L\theta_2)$$

$$F_2 = F_{10} - k_1(r_{p2}\theta_P - r_L\theta_2)$$

$$F_3 = F_{30} + k_3(r_{p1}\theta_P - r_d\theta_1)$$

$$F_4 = F_{30} - k_3(r_{p1}\theta_P - r_d\theta_1)$$



Dynamic Model of ECP Emulator

- **State Space Model** (状态空间模型)

$$\dot{x} = Ax + Bu + B_L T_L$$
$$y = Cx$$

$$x = \begin{bmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{bmatrix}, \quad u = T_D, \quad y = \theta_2$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k/g_r^2}{J_d^*} & -\frac{c_1^*}{J_d^*} & \frac{k/g_r}{J_d^*} & \frac{c_{12}/g_r}{J_d^*} \\ 0 & 0 & 0 & 1 \\ \frac{k/g_r}{J_L} & \frac{c_{12}/g_r}{J_L} & -\frac{k}{J_L} & -\frac{c_2^*}{J_L} \end{bmatrix}, B = \begin{bmatrix} 0 & 1/J_d^* & 0 & 0 \end{bmatrix}^T, \quad C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$$

$$B_L = \begin{bmatrix} 0 & 0 & 0 & -1/J_L \end{bmatrix}^T$$

$$J_d^* = J_d + J_p / g_d^2,$$

$$g_d = \frac{r_{p1}}{r_d}, g_r = \frac{r_l}{r_{p2}} \frac{r_{p1}}{r_d},$$

$$k = 2k_L r_l^2,$$

$$c_{12} = 2c_L r_l^2$$

$$c_1^* = c_1 + c_{12} / g_r^2$$

$$c_2^* = c_2 + c_{12}$$

Dynamic Model of ECP Emulator

- **Transfer Functions via Analytical Modeling** (通过分析建模的传递函数)

$$\theta_2(s)=\underbrace{\frac{g_r\left(k+c_{12}s\right)}{D(s)}}_{G(s)}T_D(s)+\underbrace{\frac{-\left(g_r^2J_d^*s^2+g_r^2c_1^*s+k\right)}{D(s)}}_{G_{TL}(s)}T_L(s)$$

where

$$D(s)=g_r^2J_d^*J_ls^4+g_r^2\left[c_2^*J_d^*+c_1^*J_l\right]s^3$$
$$+\left[g_r^2J_d^*k+J_lk+g_r^2c_1^*c_2^*-c_{12}^2\right]s^2+\left(g_r^2c_1^*k+c_2^*k-2c_{12}k\right)s$$

- **Parameter Estimation** (参数估计)

$$J_d^*=,\quad J_l=,\quad g_r=,\quad k=,\quad c_1=,\quad c_2=,\quad c_{12}\approx 0$$

Dynamic Model of ECP Emulator

- **Open Loop Transfer Function** (开环传递函数)

$$G(s)=\frac{\theta_2(s)}{T_D(s)}=\frac{b_0}{s^4+a_3s^3+a_2s^2+a_1s}=\frac{b_0}{s(s^3+a_3s^2+a_2s+a_1)}$$

Not Stable, cannot perform frequency tests!
(不稳定，不能进行频域测试！)

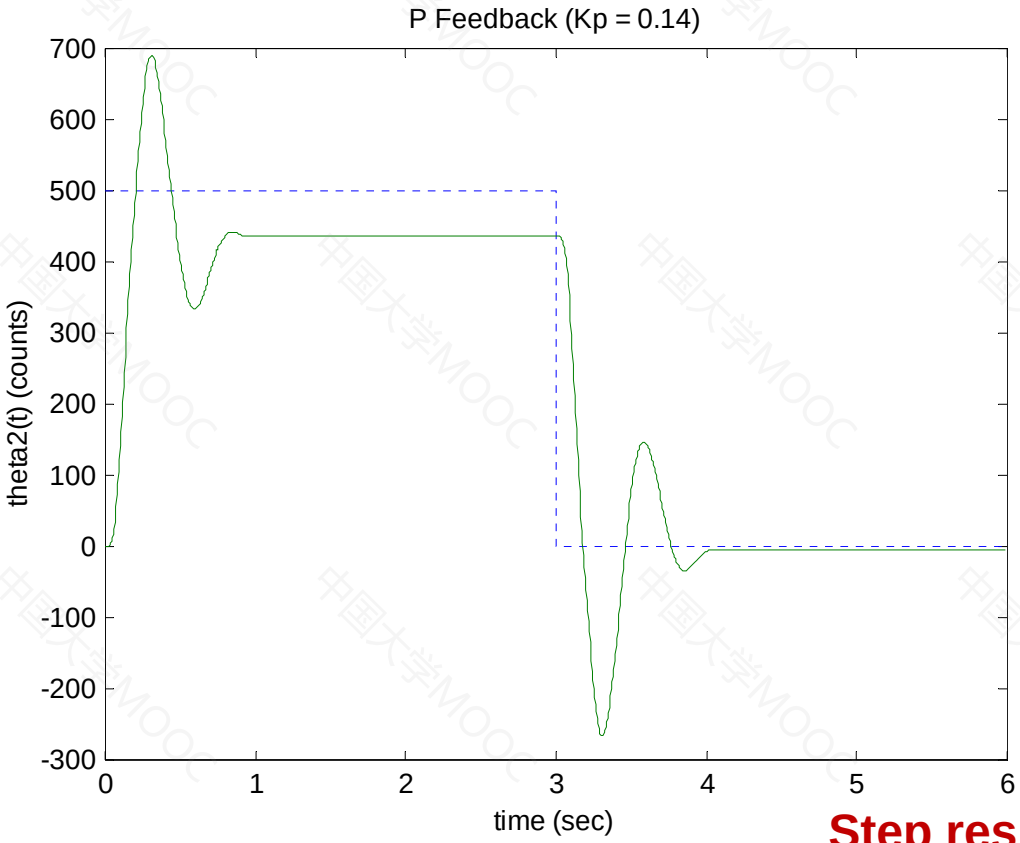
- **Closed-loop Transfer Function with P Feedback**
(比例反馈的闭环传递函数)

$$T_D(s)=-K_P\left(\theta_2(s)-\theta_{2d}(s)\right)=-K_P\theta_2(s)+K_P\theta_{2d}(s)$$

$$\Rightarrow$$

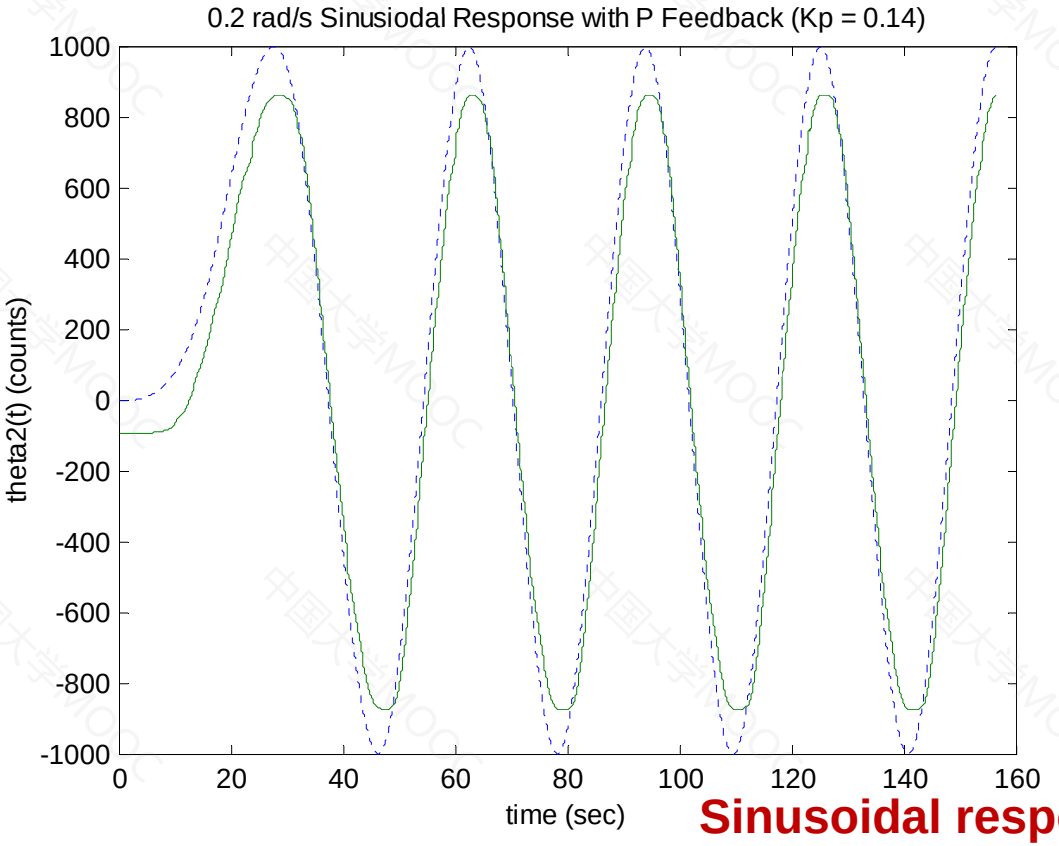
$$G_{CL}(s)=\frac{\theta_2(s)}{\theta_{2d}(s)}=\frac{K_Pb_0}{s^4+a_3s^3+a_2s^2+a_1s+K_Pb_0}$$

Response of ECP Emulator



Step response with P-feedback

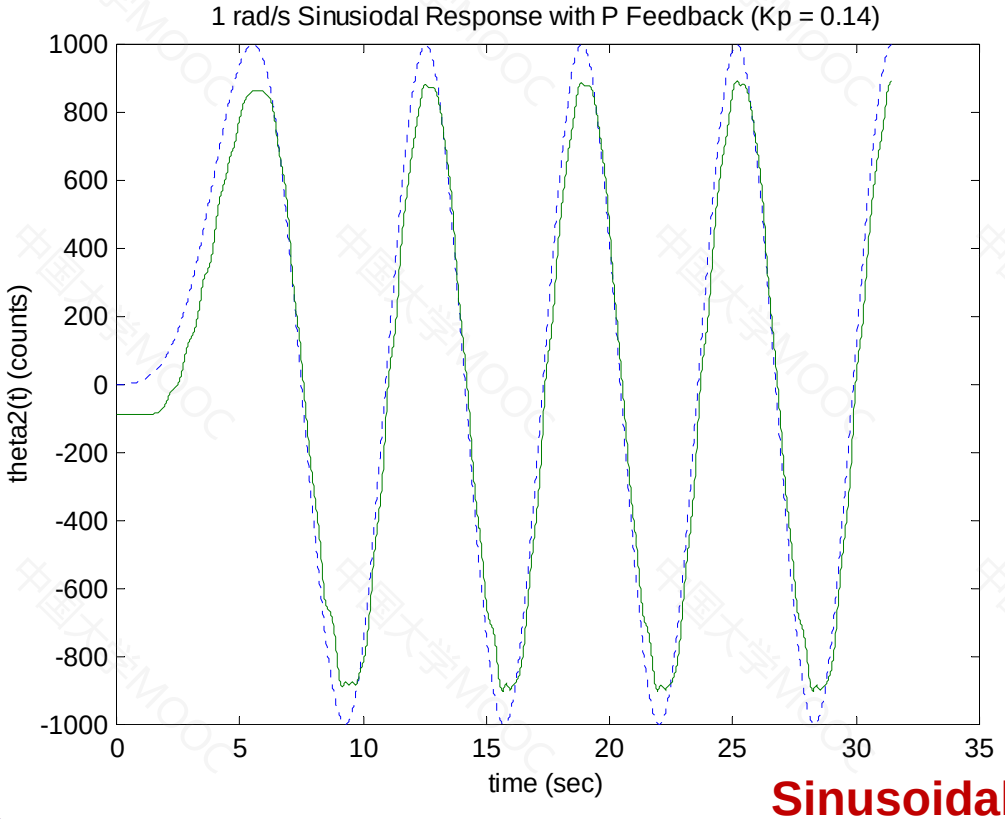
Response of ECP Emulator



Sinusoidal response with P-feedback

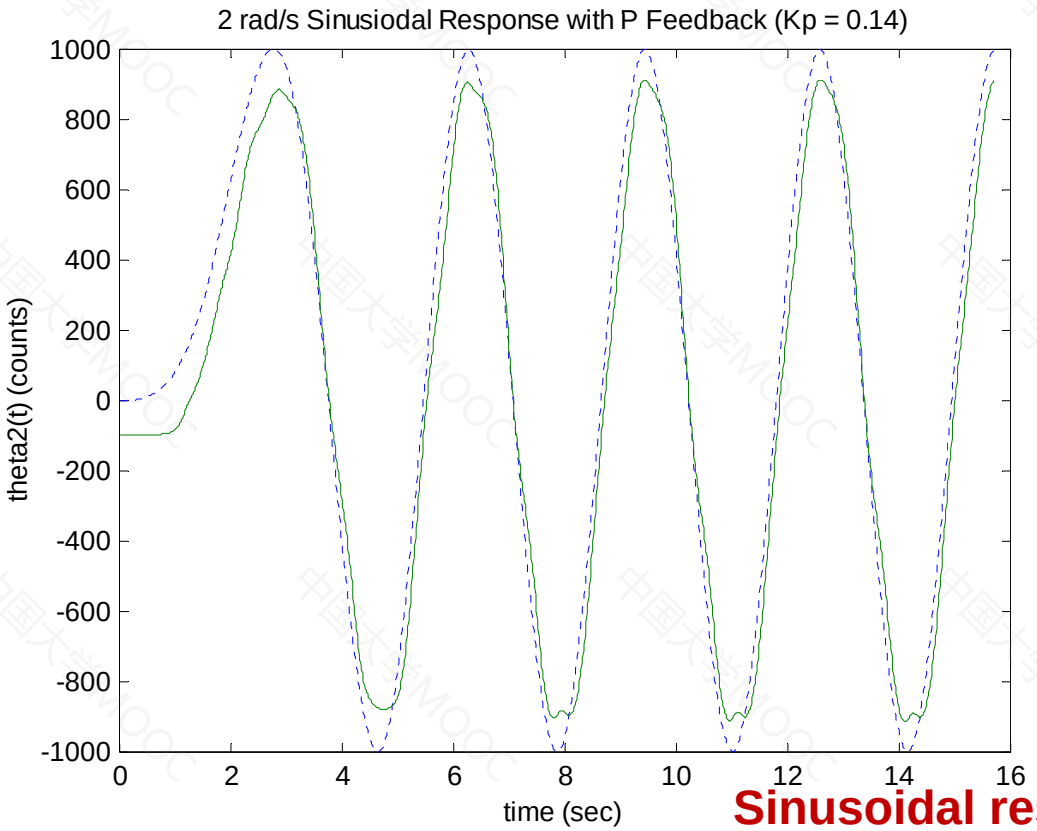
(比例反馈的正弦响应结果)

Response of ECP Emulator



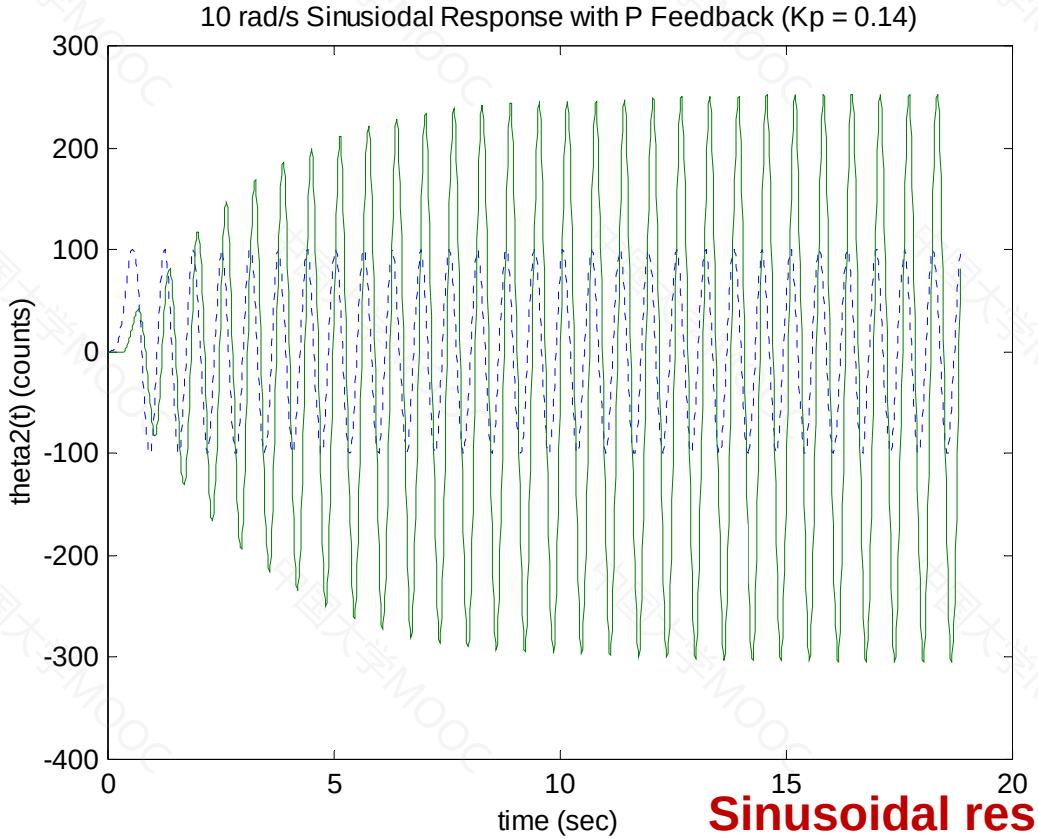
Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

Response of ECP Emulator



Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

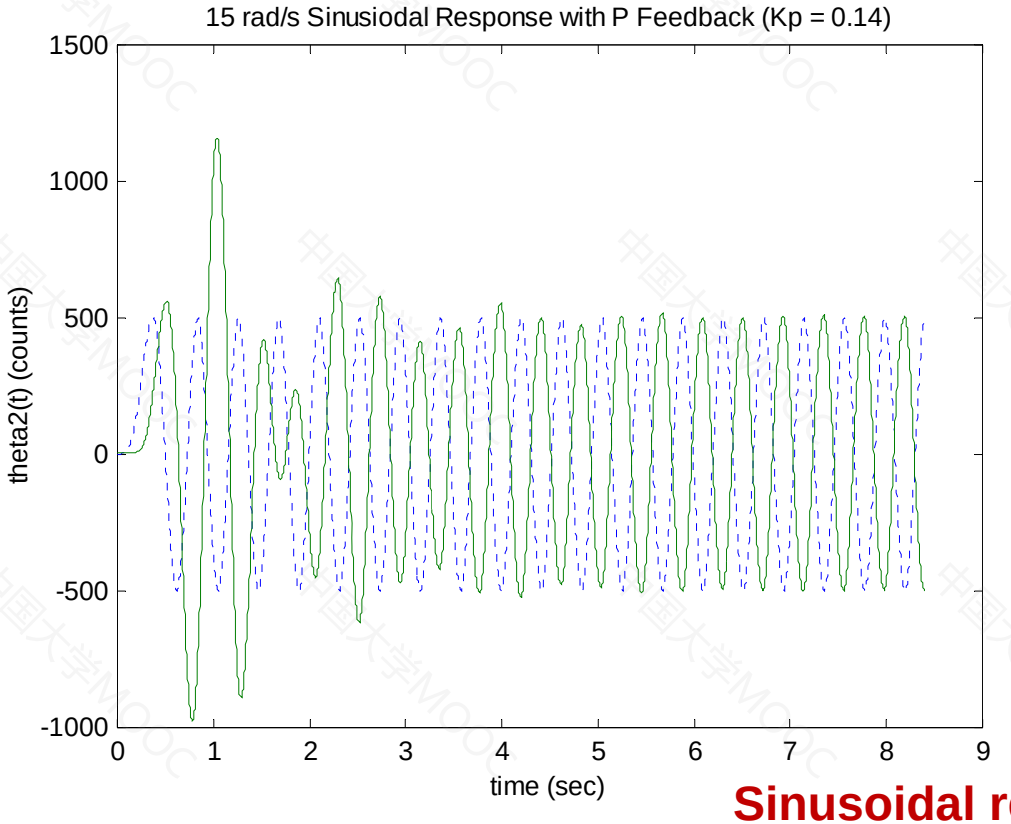
Response of ECP Emulator



Sinusoidal response with P-feedback

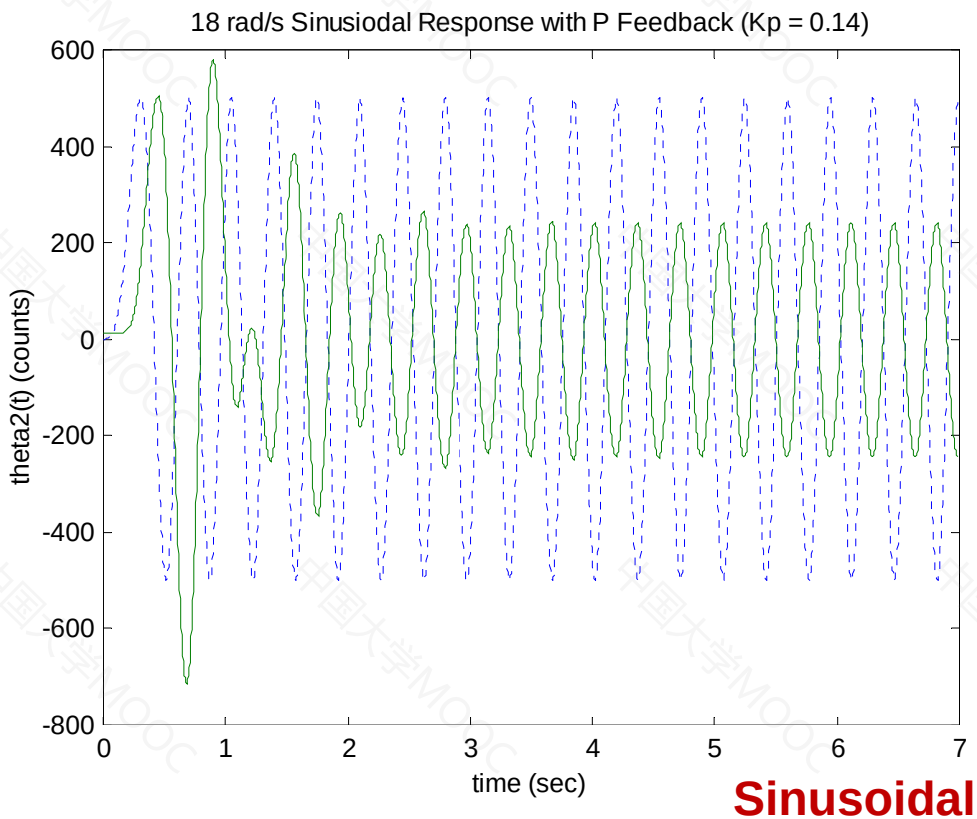
(比例反馈的正弦响应结果)

Response of ECP Emulator



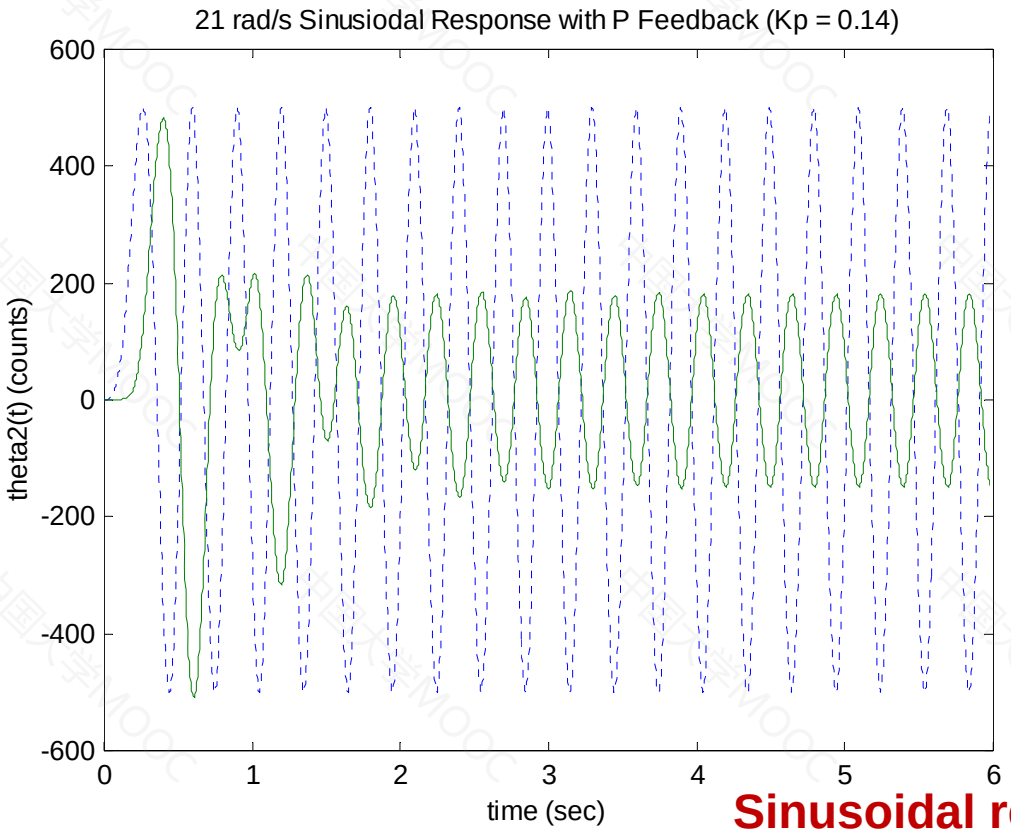
Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

Response of ECP Emulator



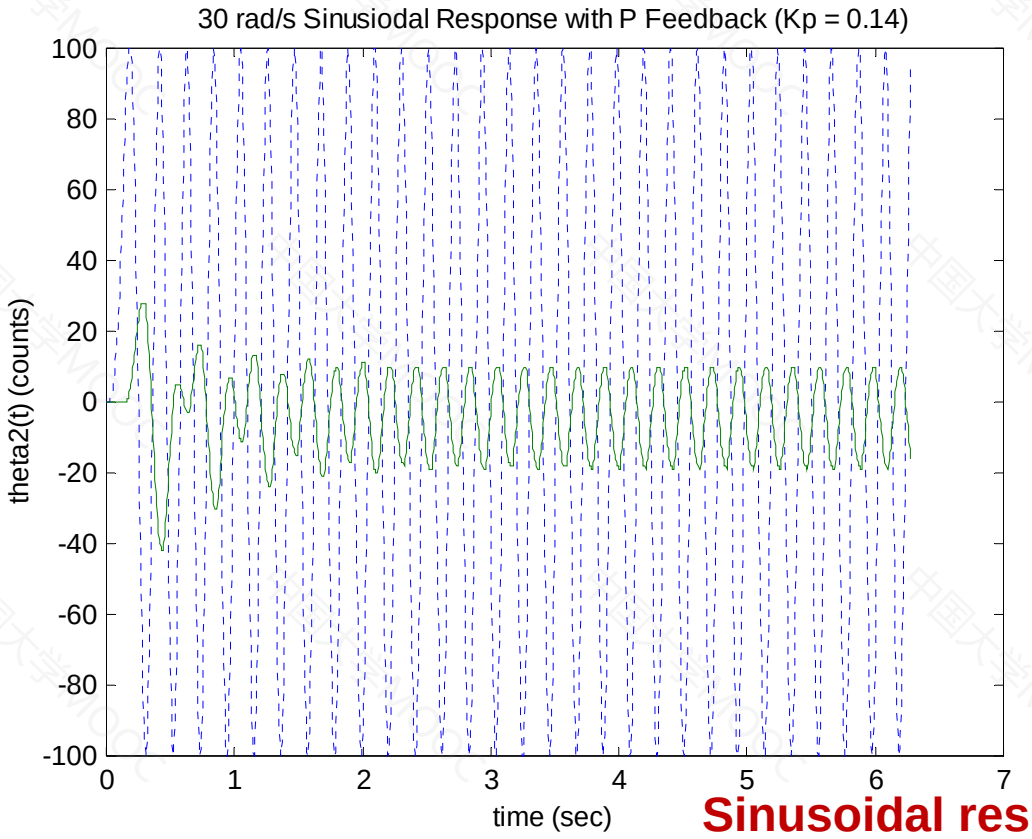
Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

Response of ECP Emulator



Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

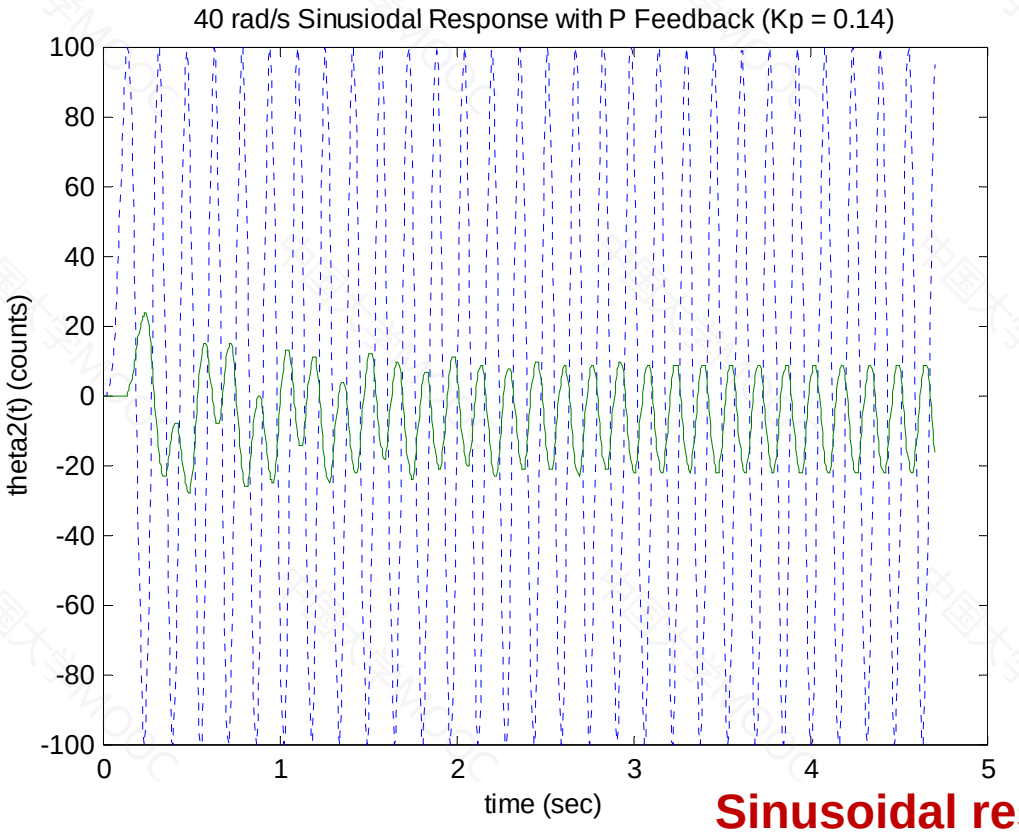
Response of ECP Emulator



Sinusoidal response with P-feedback

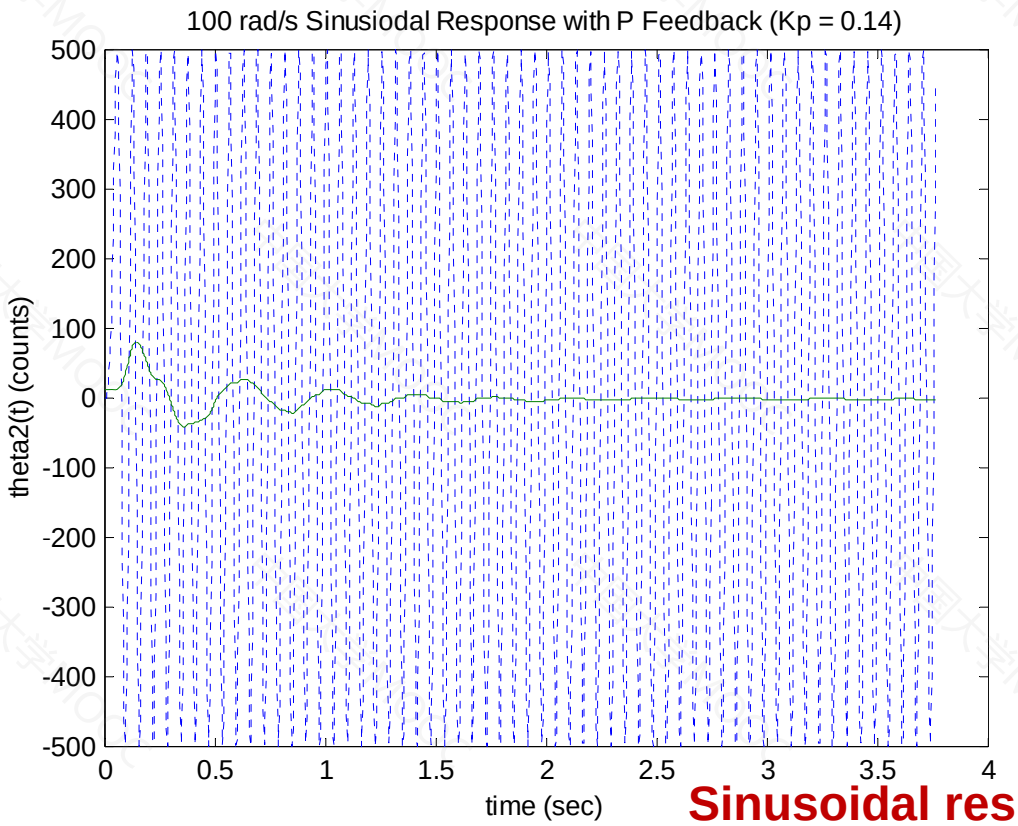
(比例反馈的正弦响应结果)

Response of ECP Emulator



Sinusoidal response with P-feedback
(比例反馈的正弦响应结果)

Response of ECP Emulator

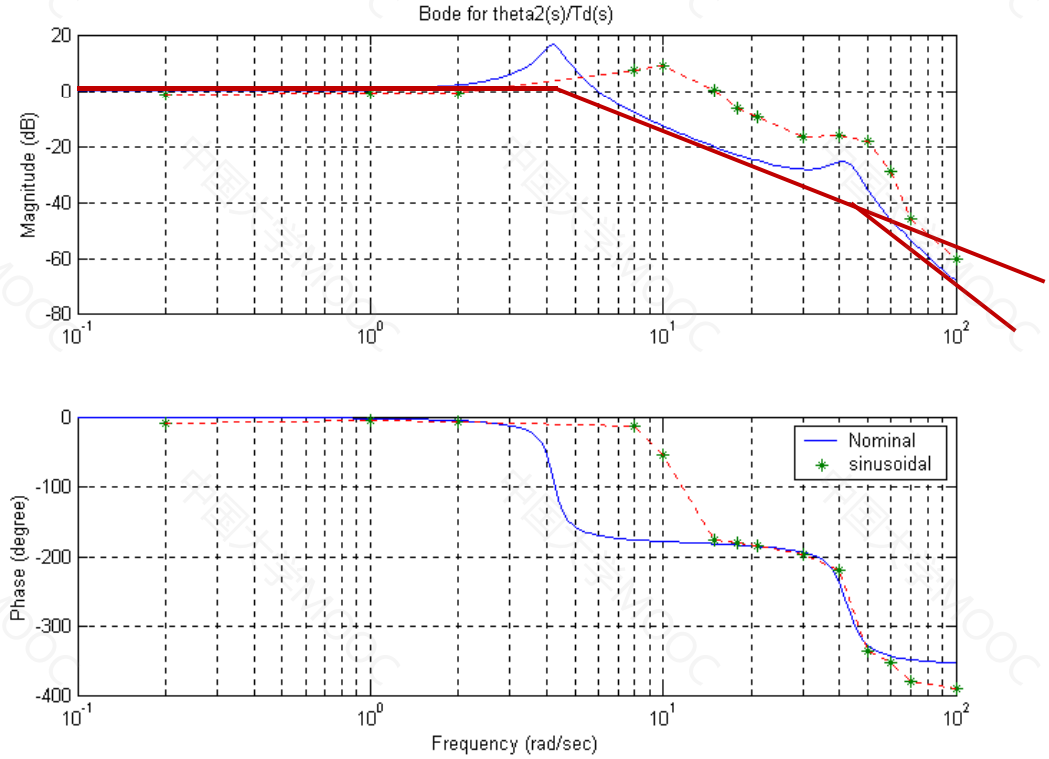


Sinusoidal response with P-feedback

(比例反馈的正弦响应结果)

Experimental Frequency Response

$$G_{CL}(s)=\frac{\omega_1^2}{s^2+2\zeta_1\omega_1 s+\omega_1^2}\cdot\frac{\omega_2^2}{s^2+2\zeta_2\omega_2 s+\omega_2^2}=\frac{31120}{s^4+9s^3+1787s^2+1185s+31120}$$



$\omega_1 = 4.2$

$\zeta_1 = 0.07$

$\omega_2 = 42$

$\zeta_2 = 0.1$

$\hat{\omega}_1 \approx 12$

$\hat{\zeta}_1 \approx 0.1$

$\hat{\omega}_2 \approx 50$

$\hat{\zeta}_2 \approx 0.1$

Frequency Response via System ID Tool Box

- **Experimental Test** (实验测试)
  - Does not need to perform many sinusoidal tests at different frequencies. Instead, generate a periodic input *u(t)* that contains components of all interested frequencies. Apply this periodic input and obtain the *steady-state* periodic response *y(t)* via one experimental test (不需要很多次不同频率的正弦测试，而是生成一个周期性的包含所有感兴趣频率的输入量***u*(*t*)**，将这个周期性的输入量作用于系统，并获得稳态周期响应***y*(*t*)**)
- **Data Processing** (数据处理)
  - Obtain the Fourier Transform of input, $U(j\omega)$, and output, $Y(j\omega)$, respectively (分别获得输入输出量的傅里叶变换)
  - Obtain Frequency Response by (获得频率响应)

$$G(j\omega) = \frac{Y(j\omega)}{U(j\omega)}$$

MATLAB System ID Tool Box

- **Input Signal Generation (输入信号的生成)**

>> help idinput

IDINPUT Generates input signals for identification **(生成用于系统辨识的输入信号)**.

U = IDINPUT(N,TYPE,BAND,LEVELS)

U: The generated input signal. A column vector or a N-by-nu matrix **(生成的输入信号)**.

N: The length of the input **(输入量的长度)**.

N = [N Nu] gives a N-by-Nu input (Nu input channels) **(N×Nu输入向量)**.

N = [P Nu M] gives a M*P-by-Nu input, periodic with period P and with M **periods (P个周期的N×Nu输入向量)**.

Default values are Nu = 1 and M =1 **(默认Nu=1,M=1)**;

Freq. Resolution **(频率分辨率)**: $\frac{1}{N} f_s$ Hz

Data Length **(数据长度)**: $N \frac{1}{f_s}$ sec

TYPE: One of the following:

'RGS': Generates a Random, Gaussian Signal **(生成一个随机、高斯信号)**.

'RBS': Generates a Random, Binary Signal **(生成一个随机、二进制信号)**.

'PRBS': Generates a Pseudo-random, Binary Signal **(生成1个伪随机、二进制信号)**.

'SINE': Generates a sum-of-sinusoid signal **(生成1个正弦叠加信号)**.

Default: TYPE = 'RBS' **(默认RGB信号)**.

MATLAB System ID Tool Box

BAND: A 1 by 2 row vector that defines the frequency band for the input‘s frequency contents **(1×2行向量定义了输入信号的频域带宽)**.

For the 'RS', 'RBS' and 'SINE' cases BAND = [LFR,HFR], where LFR and HFR are the lower and upper limits of the passband, expressed in fractions of the Nyquist frequency (thus always numbers between 0 and 1).

For the 'PRBS' case BAND = [0,B], where B is such that the signal is constant over intervals of length 1/B (the Clock Period).

Default: BAND =[0 1].

LEVELS = [MI, MA]: A 1 by 2 row vector, defining the input levels **(1×2行向量，定义输入量的幅值)**.

For 'RBS', 'PRBS', and 'SINE', the levels are adjusted so that the input signal always is between MI and MA.

For the 'RGS' case, MI is the signal's mean value minus one standard deviation and MA is the signal's mean plus one standard deviation.

Default LEVELS = [-1 1].

In the 'PRBS' case, if M > 1, the length of the data sequence and the period is adjusted so that always an integer number of maximum length PRBS periods are obtained. If M = 1 the period is chosen so that it becomes longer than P = N. In the multiinput case the signals are maximally shifted. This means that P/Nu is an upper bound for the model orders that can be used to identify systems excited by such a signal.

MATLAB System ID Tool Box

In the 'SINE' case, the sinusoids are chosen from the frequency grid freq = $\sum_{i=1}^{N} \sin(\omega_i t + \phi_i)$ 2*pi*[1:Grid_Skip:fix(P/2)]/P intersected with pi*[BAND(1) BAND(2)]. For multi-input signals, the different inputs use different frequencies from this grid. An integer number of full periods is always delivered. The selected frequencies are obtained as [U,FREQS] = IDINPUT(....), where row ku of FREQS contains the frequencies of input number ku.

The resulting signal is affected by a 5th input argument SINEDATA **(最终信号受第5个参数SINEDATA影响)**:

U = IDINPUT(N,TYPE,BAND,LEVELS,SINEDATA)

where

SINEDATA= [No_of_Sinusoids, No_of_Trials, Grid_Skip],

meaning that No_of_Sinusoids are equally spread over the indicated BAND, trying No_of_Trials different, random, relative phases, until the lowest amplitude signal is found (**No_of_Sinusoids个正弦被均匀散布在指定的频域带宽中, 随机尝试No_of_Trials次不同的相对相位，直到找到最低的幅值信号)**.

Default SINEDATA = [10,10,1] **(默认SINEDATA)**;

Generation of Exciting Signals for System ID

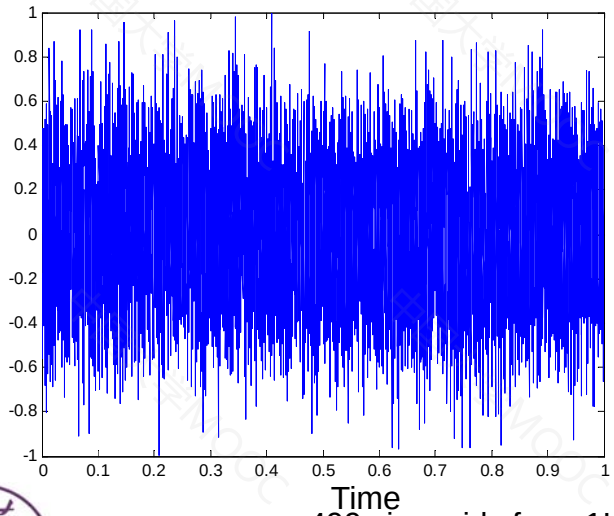
"***idinput***" command to generate a sum of 1000 sinusoids ranging from 1Hz to 5KHz with a sampling frequency of 10KHz **(产生1000个正弦叠加信号，频率从1Hz到5kHz，采样频率10KHz)**:

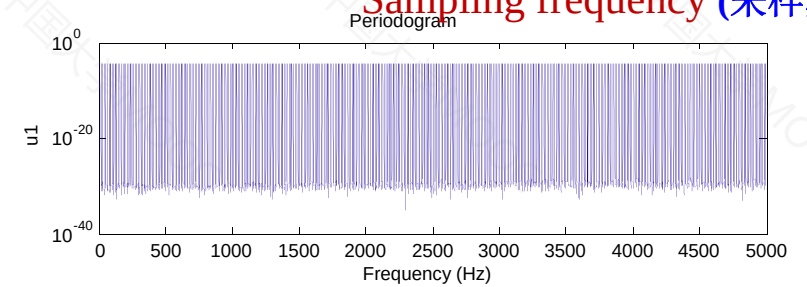
sinesum = idinput([10000 1 1],'sine', [0 1],[-1 1],[1000 50 1]);

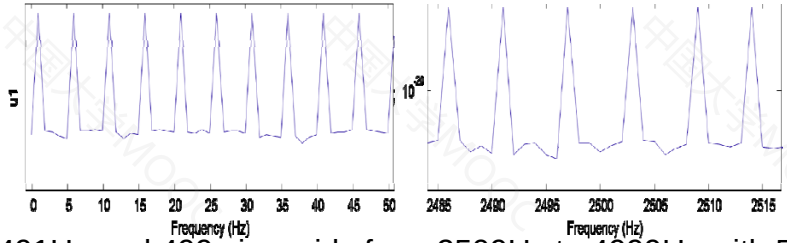
Freq. Resolution **(频率分辨率)**: $\frac{1}{N} f_s$ Hz

Data Length **(数据长度)**: $N \frac{1}{f_s}$ sec

Sampling frequency **(采样频率)**: f_s=10KHz





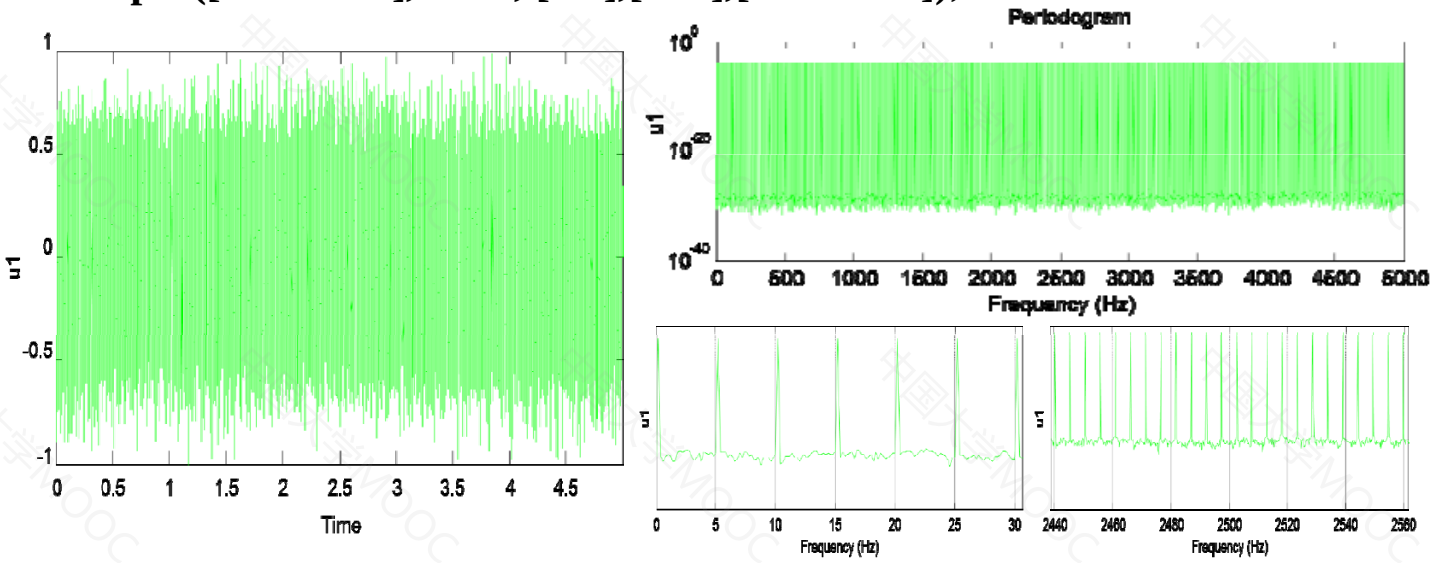


499 sinusoids from 1Hz to 2491Hz and 499 sinusoids from 2509Hz to 4999Hz with 5Hz spacing, 2 sinusoids at 2497Hz and 2503Hz with 6Hz spacing

Generation of Exciting Signals for System ID

"***idinput***" command to generate a sum of 1000 sinusoids ranging from 0.2Hz to 5KHz with a sampling frequency of 10KHz **(产生1000个正弦叠加信号，频率从0.2Hz到5kHz，采样频率10KHz) :**

sinesum = idinput([50000 1 1],'sine', [0 1],[-1 1],[1000 50 1]);

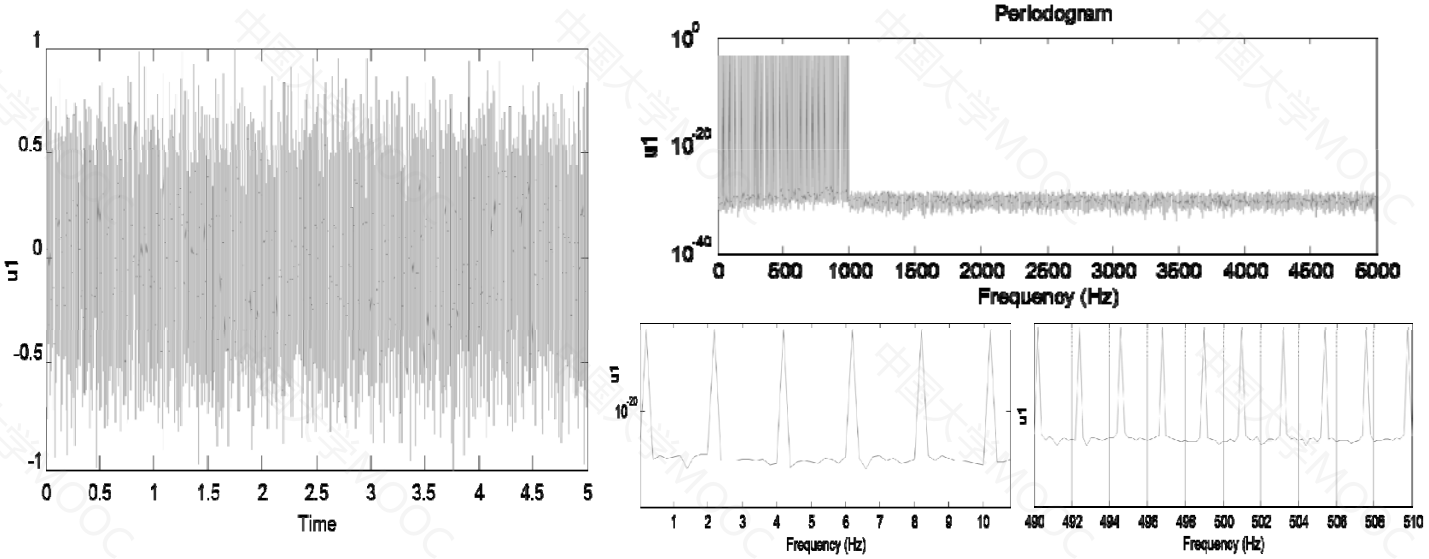


489 sinusoids from 0.2Hz to 2440.2Hz and 489 sinusoids from 2559.8Hz to 4999.8Hz with 5Hz spacing, 22 sinusoids from 2445.4Hz to 2554.6Hz with 5.2Hz spacing

Generation of Exciting Signals for System ID

"***idinput***" command to generate a sum of 500 sinusoids ranging from 0.2Hz to 1KHz with a sampling frequency of 10KHz **(产生500个正弦叠加信号，频率从0.2Hz到1kHz，采样频率10KHz)**:

sinesum = idinput([50000 1 1],'sine',[0 0.2],[-1 1],[500 50 1]);

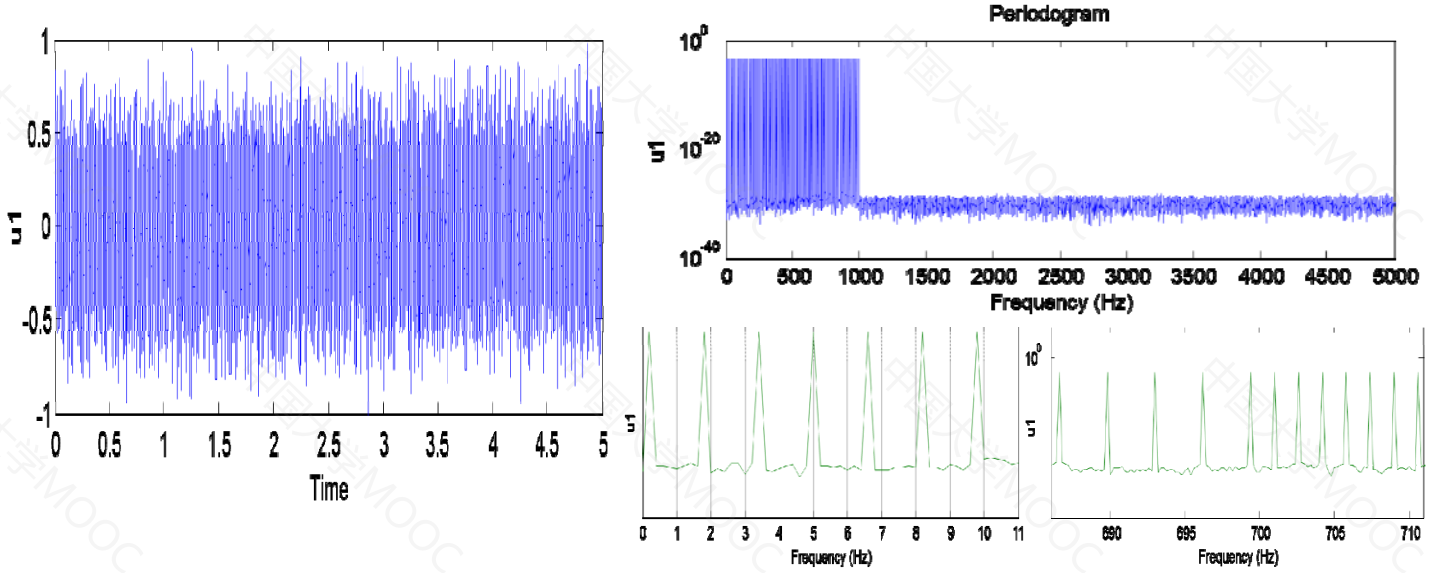


246 sinusoids from 0.2Hz to 490.2Hz and 247 sinusoids from 507.8Hz to 999.8Hz with 2Hz spacing, 7 sinusoids from 492.4Hz to 505.6Hz with 2.2Hz spacing

Generation of Exciting Signals for System ID

"***idinput***" command to generate a sum of 500 sinusoids ranging from 0.2Hz to 1KHz with a sampling frequency of 10KHz **(产生500个正弦叠加信号，频率从0.2Hz到1kHz，采样频率10KHz)**:

sinesum = idinput([50000 1 1],'sine',[0 0.2],[-1 1],[500 50 8]);



188 sinusoids from 0.2Hz to 299.4Hz and 188 sinusoids from 699.4Hz to 998.6Hz with 1.6Hz spacing, 124 sinusoids from 302.6Hz to 696.2Hz with 3.2Hz spacing

Summary

- **Experimental Frequency Response (实验频率响应)**
- **MATLAB System ID Tool Box (Matlab系统辨识工具箱)**